SCHOOL OF COMPUTER & SYSTEMS SCIENCE

MCA - Sem III

CS- 414: Theoretical Computer Science

Mid-Semester Examination

Date: 7-10-2024

Time: 9:30am – 11:00am **Max. Marks: 30**

Part-A of Total 10

Answer any FIVE of the following. Each carry TWO marks.

1) Given, firstly, the propositions **Q** and **R**, and the formula **Q** arrow **R.** Secondly, for alpha, beta belonging to the set of all strings over the union of sets N and Σ, and the production rule **alpha arrow beta** in an arbitrary grammar, **G** equal to the four tuple N, Σ, P, S. Then distinguish **Q arrow R** from **alpha arrow beta**.
2) Given two arbitrary sets L 1, L 2, and let L 3 be equal to the complement of the union of L 1 complement, and L 2 complement. $(L_1' \cup L_2')' = L_3$. Deduce the an equivalent expression equal to how L 3 is described. Students with vision have to deduce the equality using the Venn diagram what L 3 is.
3) Consider the elements a b c and d to be elements of the sets **X**, 1 2 3 be the elements of the set **Y** and a mapping p with set X as domain and set Y as the range. List all the possible types of mappings/functions/relations/ homomorphism/isomorphism that may be defined by **p**?
4) Using an example prove the set theoretic inequality – the cardinality of the union of two arbitrary sets is less than or equal to the sum of the cardinalities of the two sets. $|\mathbf{S} \cup \mathbf{T}| \leq |\mathbf{S}| + |\mathbf{T}|$.
5) For the power set of an arbitrary set say A, how can a lattice over the power set of A be obtained?
6) Given the well-formed formula wff 1 – left parenthesis Q **and** R right parenthesis infer R. $(Q \wedge R) \rightarrow R$, Determine if wff 1 is satisfiable, or a tautology or a contradiction.
7) Give the regular expression for the regular set of passwords, say L_password, defined over four sets namely, the set of lowercase letters, the set of uppercase letters, the set of digits and a set of four special characters (of your choice) such that

i) the length of the password must be less than or equal to ten, and

ii) it must begin with an uppercase or lowercase letter, and

iii) it must not end with a special character must begin with a uppercase letter

Part-B [20]

Answer any FOUR of the following six questions carrying FIVE marks each.

8) a) Give any two advantages of the Kleene's Theorem in the theory of automata. [2]

b) Design either a Mealy machine or a Moore Machine for a sensor enabled automatic door. Justify your choice of the machine. Also justify if neither of these finite state machines is possible for the given problem. [3]

9) Let **L one** be a regular set over the alphabet **Σ one**, and **L two** be a regular sets over the alphabet **Σ two**. Prove that concatenation of these two regular sets say, **L two L one** is a regular set. Further, for **w one** an element of the set **L one** and **w two** an element of the set **L two**, show that for L two L one over the alphabet Σ one union Σ two, **w** equal to **w two w one** is an element of the set **L two L one**.

10) Give a deterministic finite automata **M one** over a set including the symbols a and b, that can accept all strings ending with **aa**. Further, for the following two inputs what does the machine yield? Alternately, describe the functioning of **M one** for the following inputs: [5]

i) abababaa

ii) baabaaba

11) Define a grammar. Give the distinguishing properties of the Context Sensitive Grammar, Context Free Grammar and Regular Grammar and name the recognizers for each. [2+3]

12) Consider the following set of production rules of an arbitrary grammar G:

i) S arrow open parenthesis S close parenthesis, [S → (S)]

ii) S arrow S plus S, [S → S + S]

iii) S arrow S star S, [S → S * S]

iv) S arrow **id,** [id, written in small letters and boldened]

Give the remaining components of the G, required to generate the Language L of G. Name/describe the language L of G. Also give one property of the grammar.

13) Define any five of the following,
[1x5]

a) **string** over a nonempty set of symbols

b) **Kleene closure**

c) **instantaneous description**

d) **Sentential form**

e) **derivation tree**

f) **transition function** δ of a Push Down Automata (PDA)

End of question paper.

School of Computer and Systems Sciences
MCA – Semester III
CS-414: Theoretical Computer Science
End-Semester Examination

Date: December 9th 2024
Max. Marks – 50
Time: 10:00am to 12:30pm

Part-A of 20 marks

Answer any TEN of the following questions. Each carry two marks.

1. Let *p* and *q* be the propositional statements. What is the **negation**, **converse**, **inverse** and **contrapositive** of the implication *p implies q*?
2. For a Deterministic Pushdown Automata (DPDA) give the transition functions that depict **Push** and **Pop** stack-operations in the stack.
3. Consider a universal set U of members a, b, c, d, e, f, g, h, i, j. Let any subset of U follow the same sequence. If the member of subset A be a, b, e, g, h j and the bit string of B proper subset of U be **1101 0010 10**, then give the bit strings of complement of A and complement of A intersection B.
4. State one of the decision problems related to Regular Sets.
5. In the following basic logical equivalences for the propositional statements *p* and *q*, give the missing terms,
   a. absorption law:
      i. *p* and blank equivalent to *p*
      ii. *p* or blank equivalent to *p*
   b. idempotent law:
      i. *p* and blank equivalent to *p*
      ii. *p* or blank equivalent to *p*
6. Define a Turing Machine.
7. Let the members of a set A be 1, 2, 3, 4. Let **R** and **S** be relations on **A** defined by,
   R is a set of pairs, pair 1,1; pair 3,3; pair 1,3; pair 2,3; pair 3,2; pair 4,2 . Set R contains 6 pairs.
   S is a set of pairs, pair 1,1; pair 3,3; pair 1,3; pair 3,1; pair 2,3; pair 3,2; pair 4,2; pair 2,4. Set S contains 8 pairs.
   List the properties of **R** and **S,** as defined above?

8. Given an alphabet Σ, the set containing 0 and 1. List the first twelve strings in the lexicographic order. Let **w** an element of the set of all strings over 0 and 1, such that the length of w is 7. Give the string w if $a_1a_2a_3a_4a_5a_6$ be the prefix where each a_i is either 0 or 1 for value of i being $1 \leq i \leq 6$.
9. Consider the grammar G = (N, T, P, S), where set N includes the symbol S. The set T comprises two symbols, namely open parenthesis and closing parenthesis. P includes the rules, S arrow empty symbol, S arrow SS, and S arrow open parenthesis S closing parenthesis. Consider a string **w** an element of the set L defined by G. Draw the derivation tree for **w = open close open open close open open close close close**.
10. Suppose a function *f defined over the domain A having range B* and function *g defined over the domain B having range C*. Then prove that if *g composition f* is onto C then *g* is onto C.
11. Give an example of an ambiguous grammar, say G. Also give G′, the equivalent grammar, after removing its ambiguity.

Part – B [30]

Answer any TREE of the following. Each carry 10 marks.

12. a. Consider the alphabet Σ of members a and b and languages L one and L two represented by the regular expressions r one equal to ab plus aab whole Kleene closure and r two equals aa concatenation a+b Kleene closure concatenation bb. Give the Finite Automata M to recognize all strings of the language L one concatenation L two. Is M deterministic or nondeterministic? 6 marks

    b. Consider the statement - *There is a relation between Recursively Enumerable sets and NP class of problems*. Discuss if the statement is **True** or **False**. 4 marks
13. a. Describe the steps comprising the proof of a mathematical statement by the Principle of Mathematical Induction by using either an example or the statement of the principle. Five marks

    b. Give the grammar for the set of palindromes of the form w c reverse of w, where w is any string over the alphabet comprising symbols 0 and 1. Does L contain the empty word? 5 marks
14. a. Let the set D_{20} comprise the set of all positive divisors of 20. Then give the digraph and Hasse diagram of D_{20}. 4 marks

b. State the Pumping Lemma for Context Free Languages. Describe the significance of the constant, and the conditions stated in the lemma with respect to a string **w**. 6 marks

15. a. Let G the quadruple N, Σ, P, S be a grammar with the production rules of the form α arrow β. Name the types of grammars and their properties. 4 marks

   b. Design a Turing Machine over **the alphabet with members a, b and c** which for inputs m number of a's followed by n number of b's, given m than or equal to 1 and n greater than or equal to 1, will output the string m number of a's followed by one c followed by n number of b's. 6 marks

16. Design a Pushdown Automata over the alphabet comprising symbols 0 and 1 that may accept the set of all strings of m ones followed by n number of zeros where m is greater than or equal to n. Demonstrate the functioning of M with input strings x one equal to 111000, and x two equal to1000. Marks [5+5]